

CON 20
1

Orchestrating containers on AWS

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Agend

a

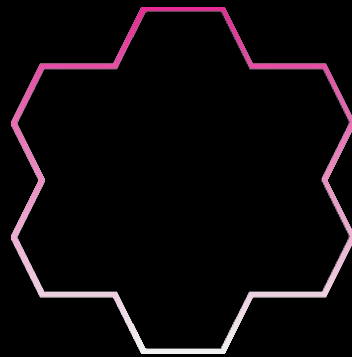
Microservices and
containers

AWS container services

Datree story and live

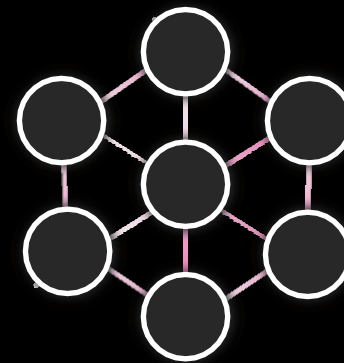
demo

When the impact of change is small, release velocity can increase



Monolith

Does everything

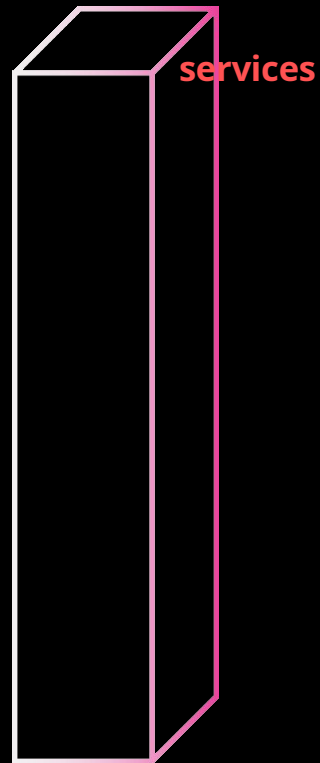
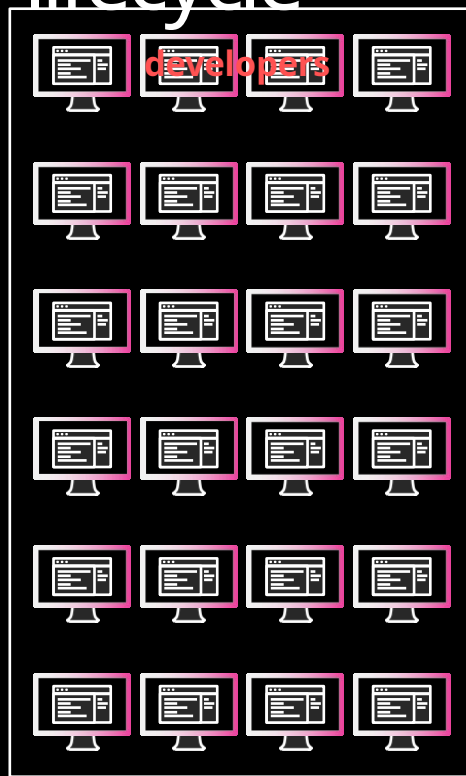


Microservice

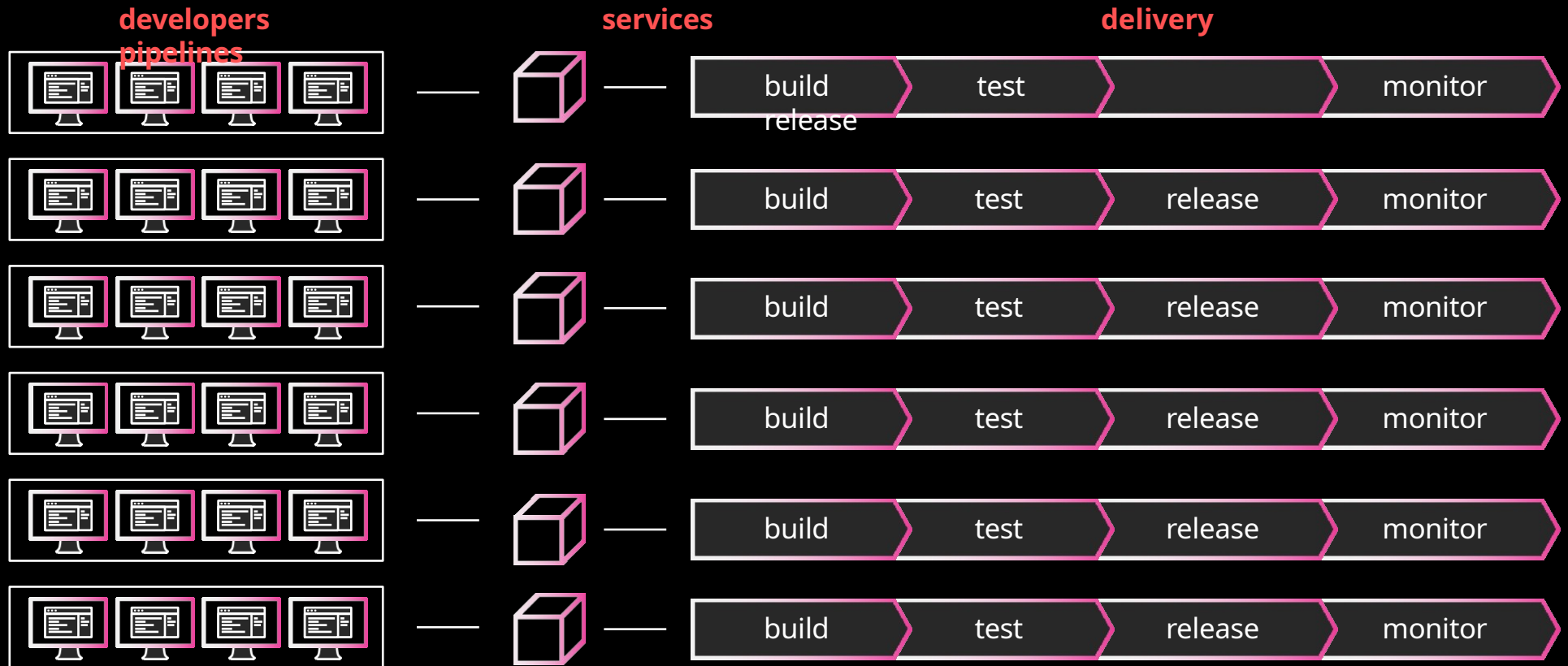
s

Does one thing

Monolith development lifecycle

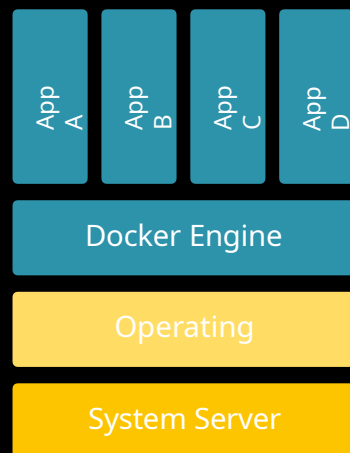


Microservice development lifecycle



Containers and Docker

A container is a standard **unit of software** that packages up **code and all its dependencies** so the application **runs quickly and reliably** from one computing environment to another.¹



¹ <https://www.docker.com/resources/what-container>

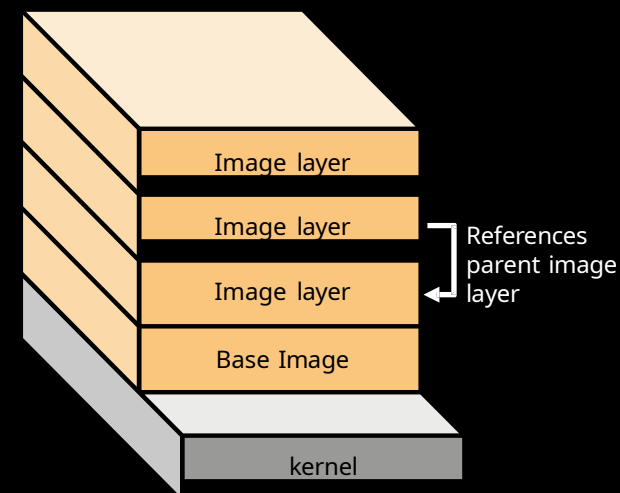
Docker Image

Used to launch
container

Instructions documented in
Dockerfile

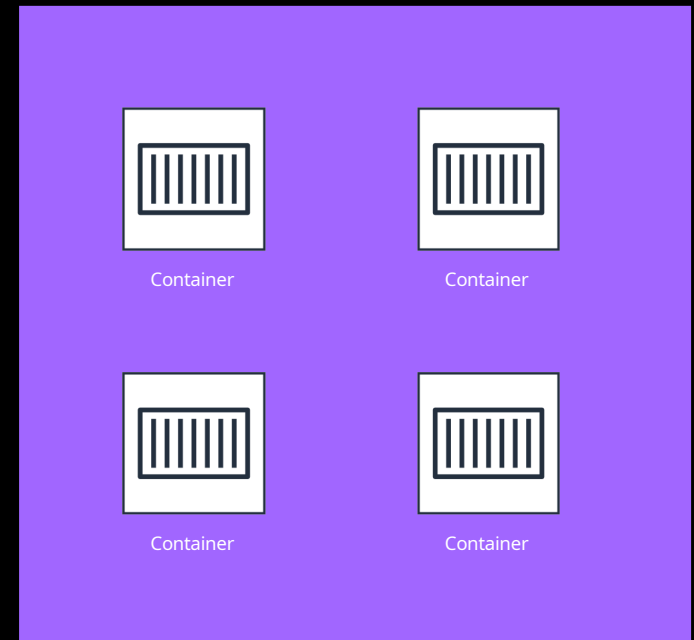
Merge layers into single
image

Read-only
template

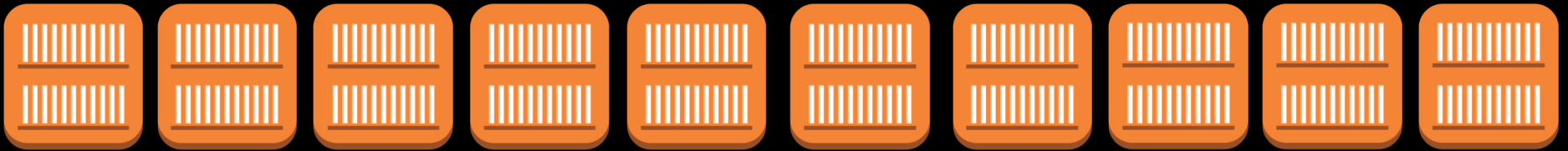


Containers and Microservices

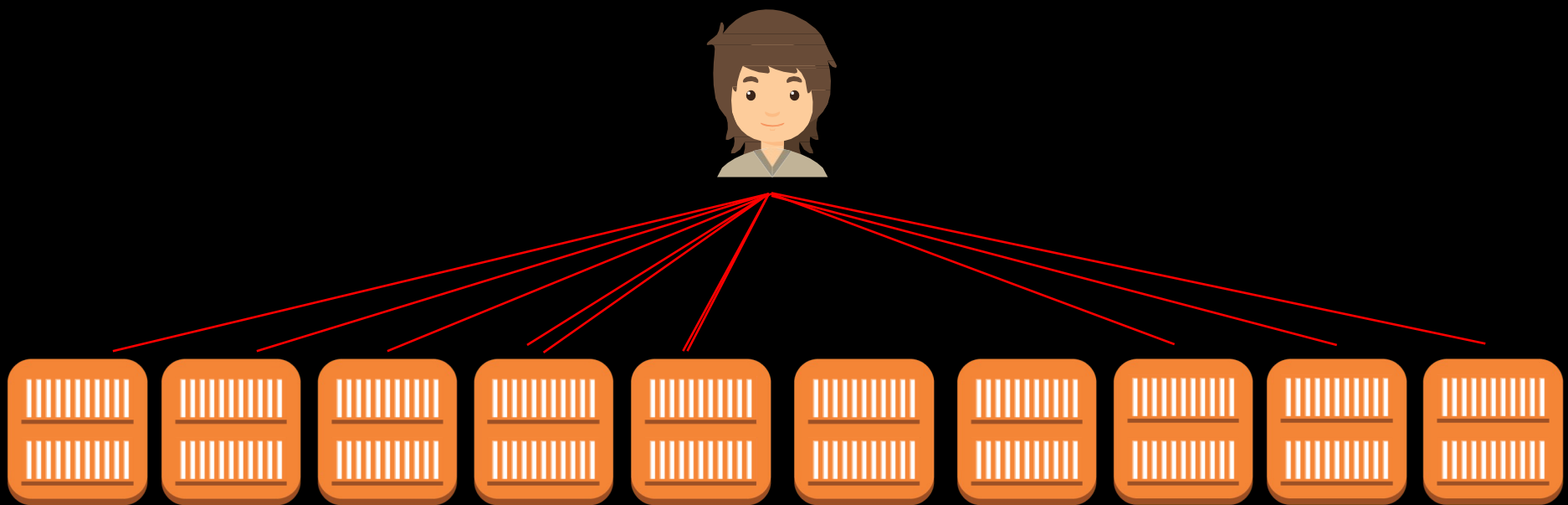
- Do one thing, really well
- Any app, any language
- Isolated execution environment
- Test and deploy same artifact
- Faster startup



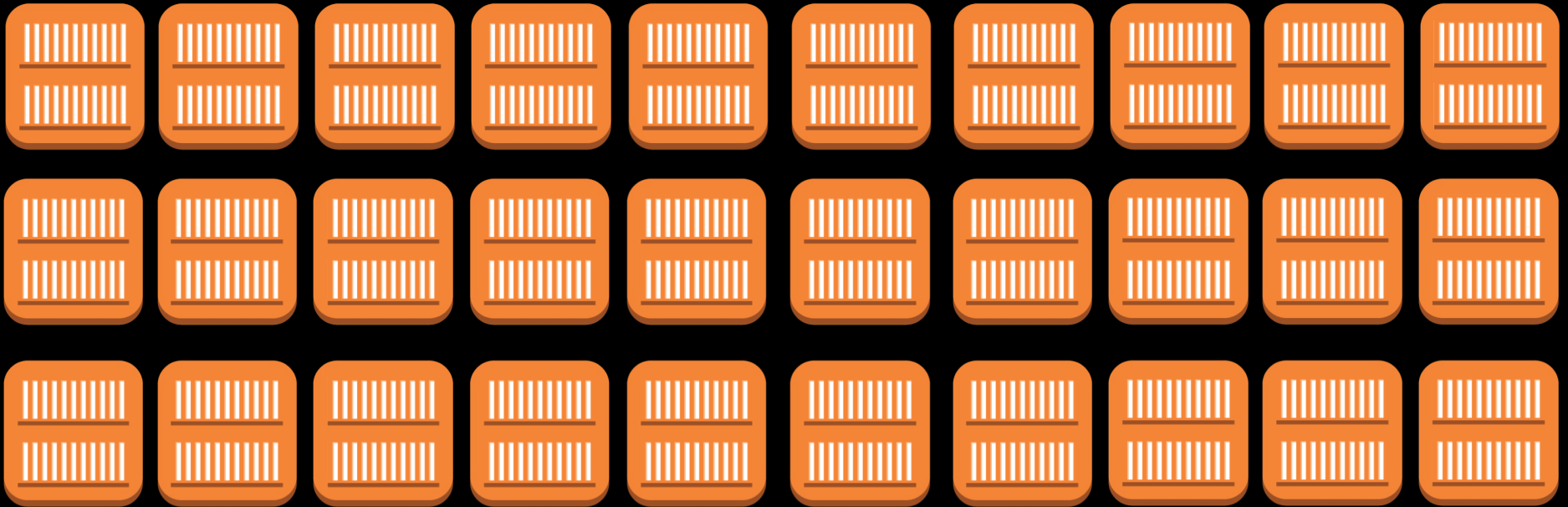
Containers have become the standard for how to ship and run
your application in the cloud



Manually downloading and launching containers by hand is inefficient and error prone



Container orchestration



AWS container services

AWS Container Services landscape

MANAGEMENT

Deployment, scheduling, scaling
& management of containerized
applications



HOSTING

Where the containers run



IMAGE REGISTRY

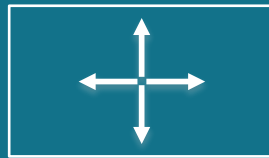
Container image
repository



Kubernetes



Open source container
management platform

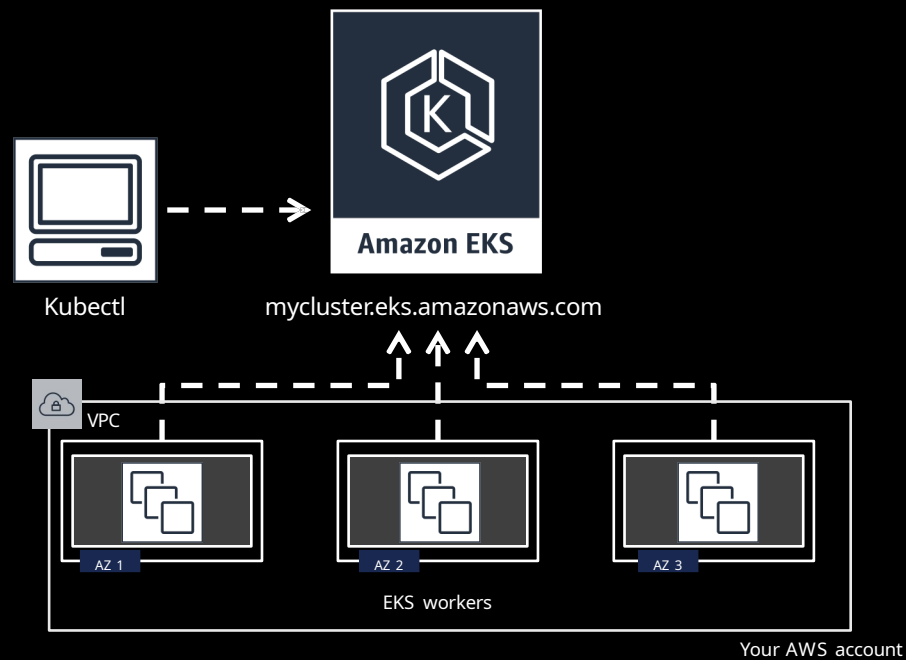


Helps you run
containers at scale



Gives you primitives
for building
modern applications

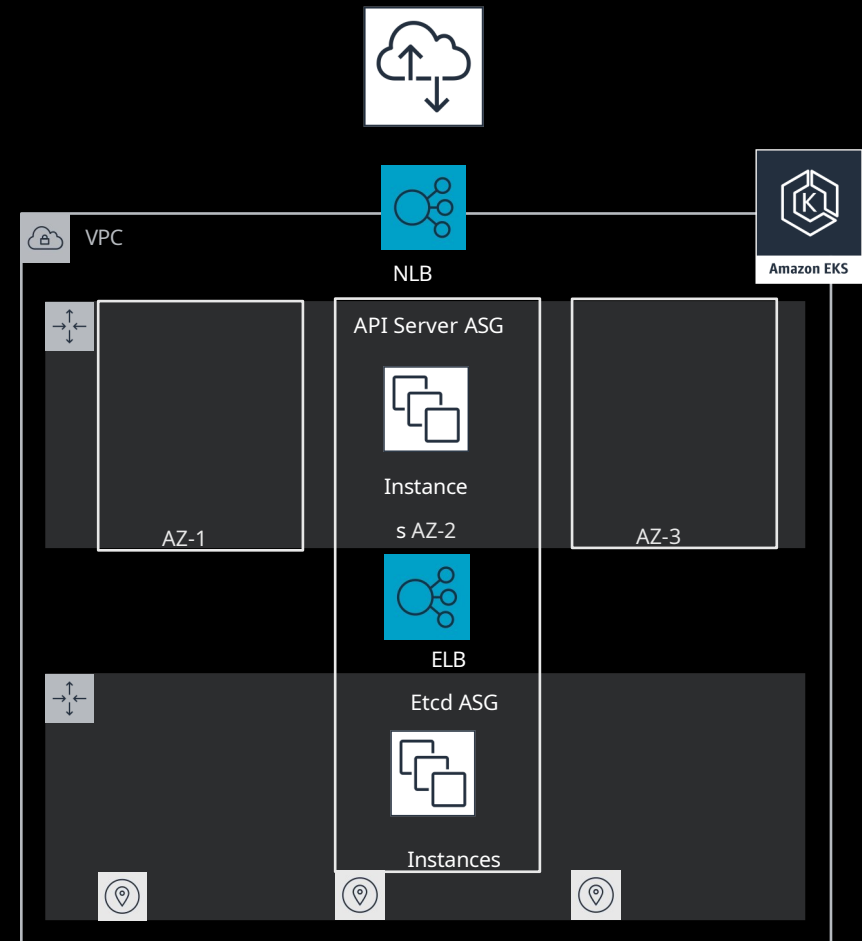
Amazon EKS architecture



Kubernetes control plane

Highly available and single tenant infrastructure

All “native AWS” components
Fronted by an NLB



Amazon EKS features

Certified conformant

Integration with Elastic Load

Balancing Managed updates

IAM authentication

Amazon Elastic Container Service



EC
S

Highly Scalable ,
Highly Performant
Container
Management
System

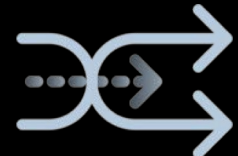
A managed
platform



Cluster
Manageme
nt

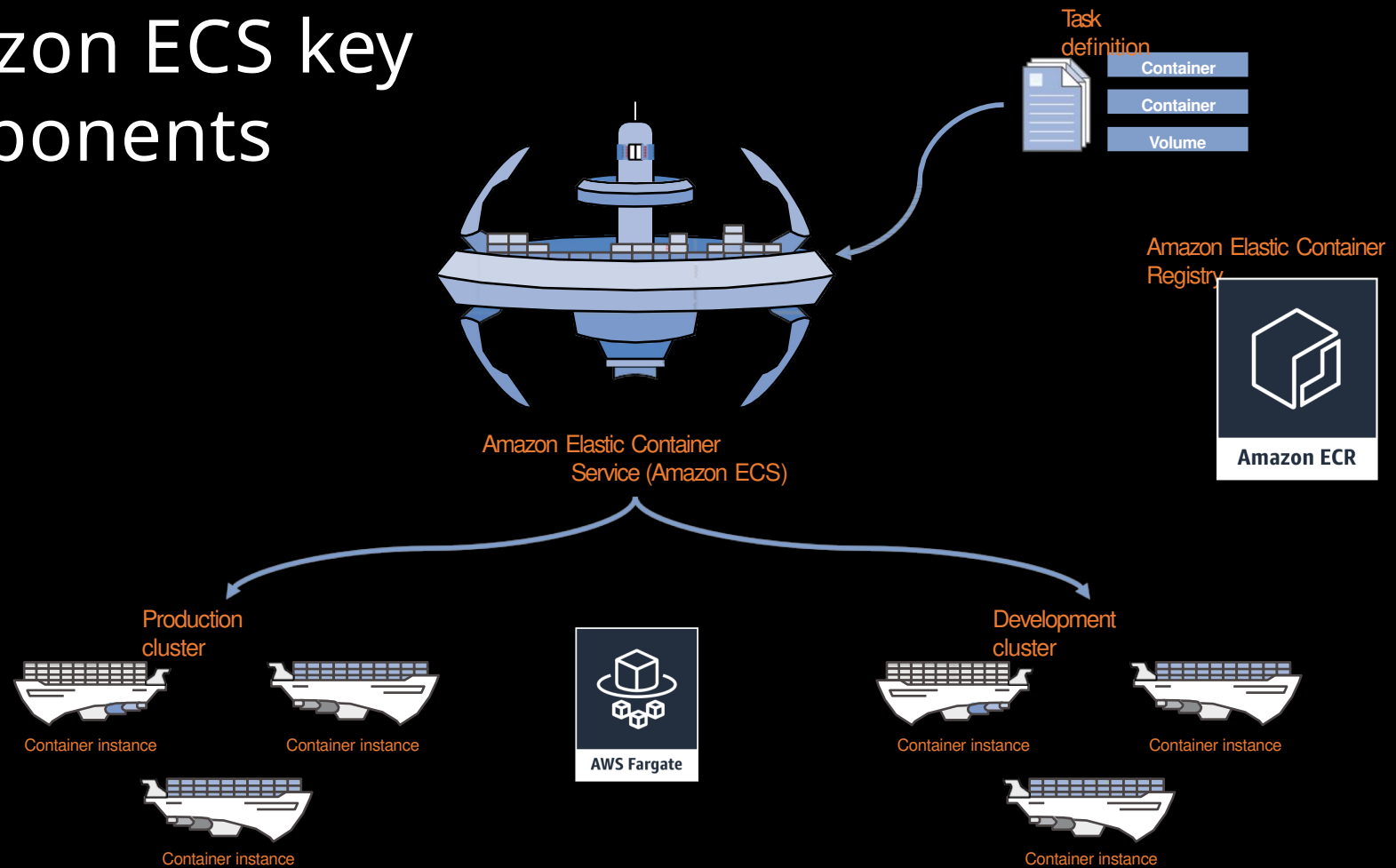


Container
Orchestration
& Placement



Deep
AWS
Integratio
n

Amazon ECS key components



Amazon ECS features

Integration with Elastic Load Balancing

Service Discovery with AWS

CloudMap Task level IAM support

Blue/Green Deployments with AWS

CodeDeploy Windows Containers

Compatibility

EC2

hosting

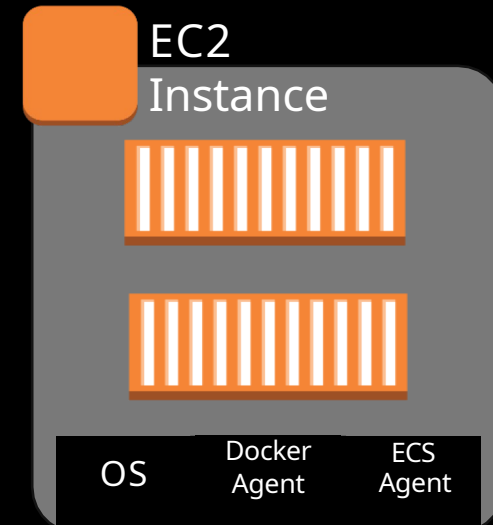
Choose your instance type

Connect to the instance

Persistent EBS storage

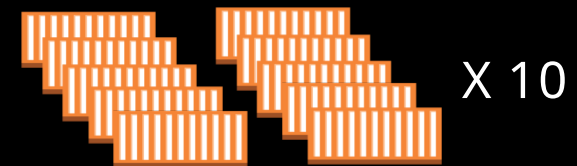
Custom kernel

modules





“Just launch 10 copies of
my container
distributed across three
availability zones and
connect them to this
load balancer”



Shimon Tolts CTO & Co-Founder Datree



Agend

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1

About me &
datree.io

4

How do we deploy
(technical)

2

What is Fargate
TCO?

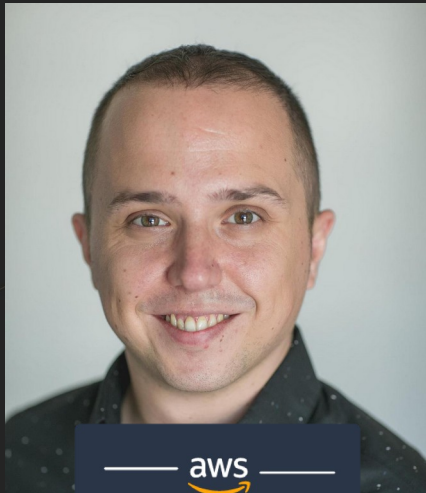
5

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Why did we
choose to use
Fargate

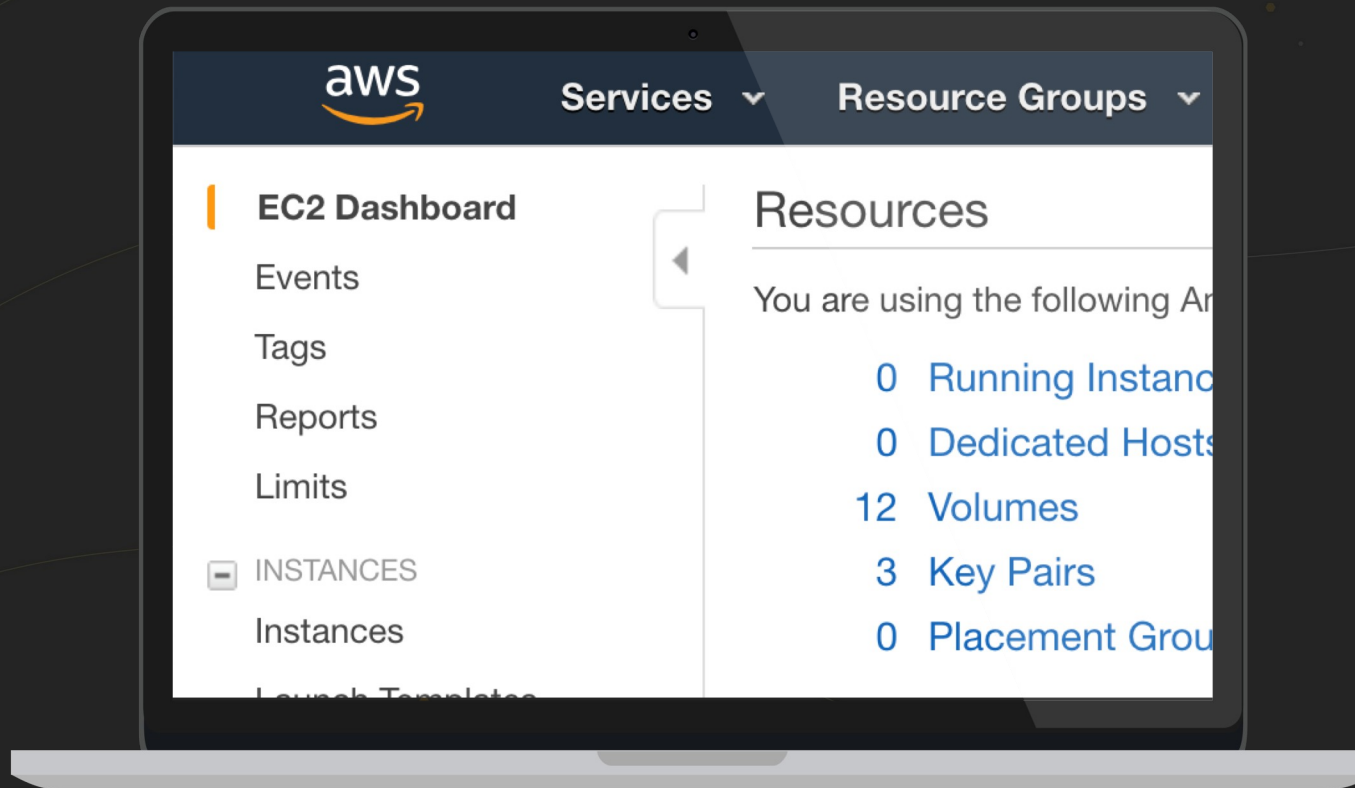
About me



— aws —
HEROES

```
Shimon Tolts {  
  age: 30,  
  title: "CTO & Co-Founder @ datree.io",  
  misc: ["AWS Community Hero", "Gamer"]  
}
```

0 EC2 instances @ datree.io





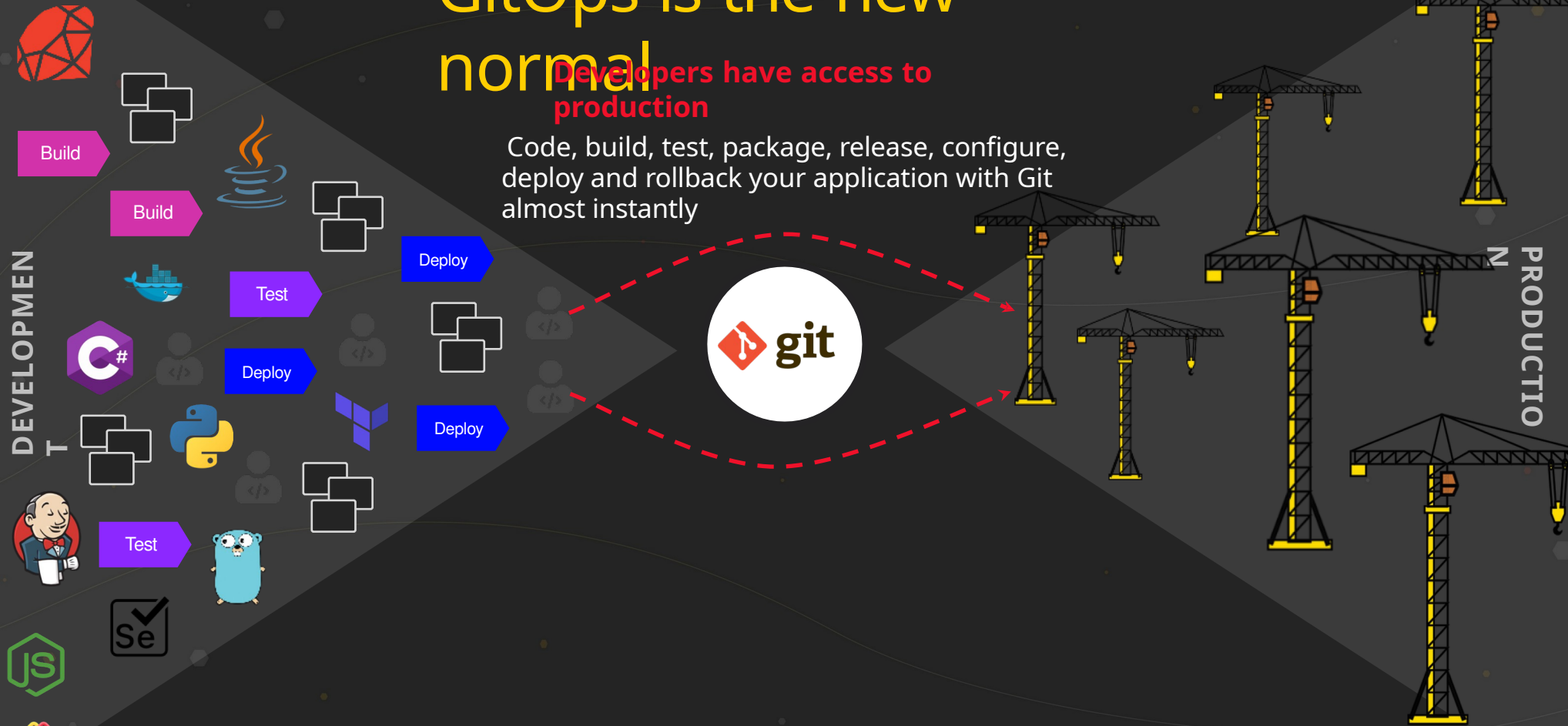
GitOps

Datree is a Policy Enforcement
Platform for confident and
compliant code.

GitOps is the new normal

Developers have access to production

Code, build, test, package, release, configure, deploy and rollback your application with Git almost instantly



Datree connects with GitHub



Datree connects with **GitHub** to provide automatic policy compliance checks and insights for every code commit and pull request.

Datree connects with GitHub

Set Code Policies

Create custom policies or choose from recommended defaults, and choose where they should be enforced.



Separate
secret
credentials
from source
code



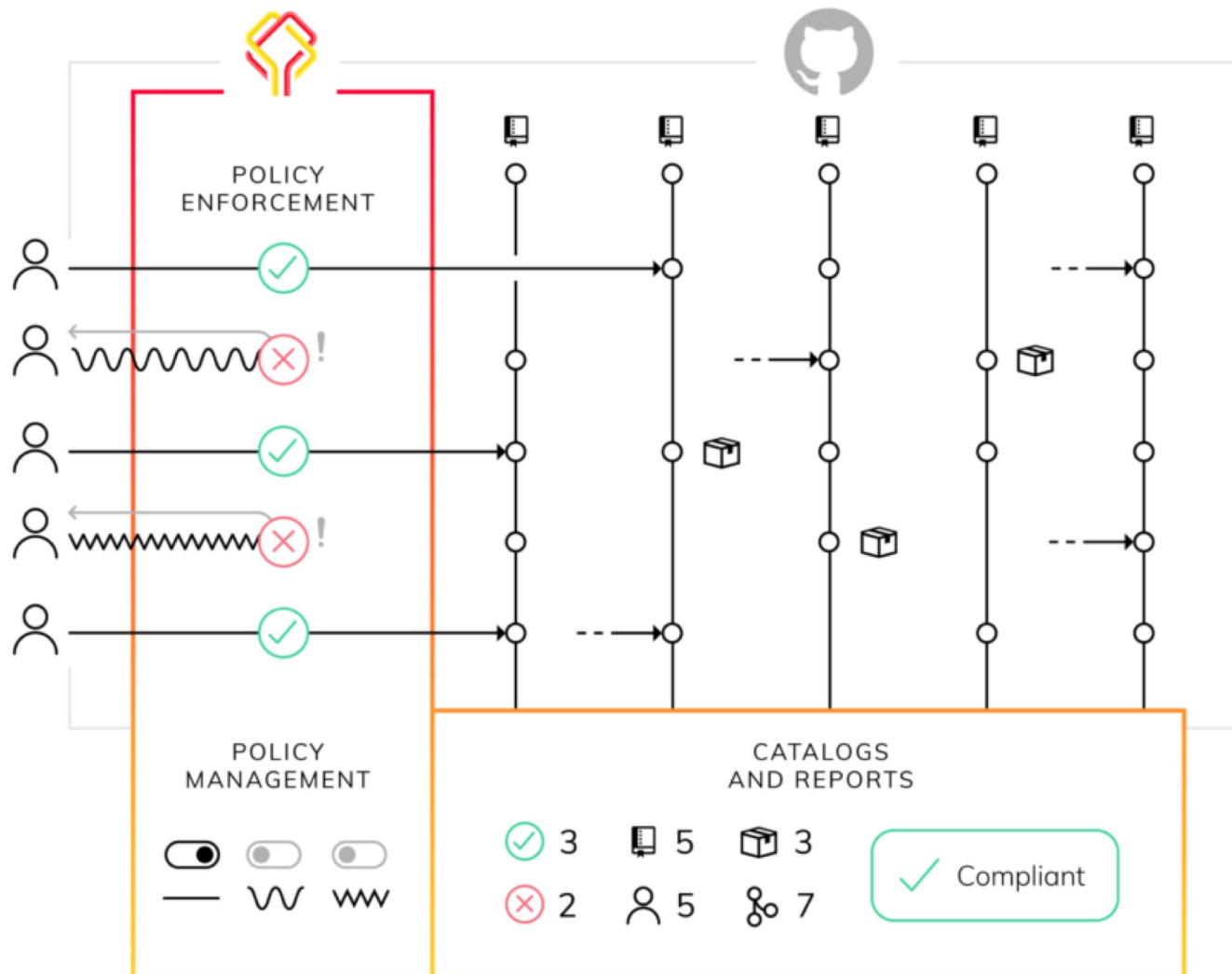
Include .gitignore in every
project



Link pull
request title to
a Jira ticket



Create
custom
policies...



Apples and oranges

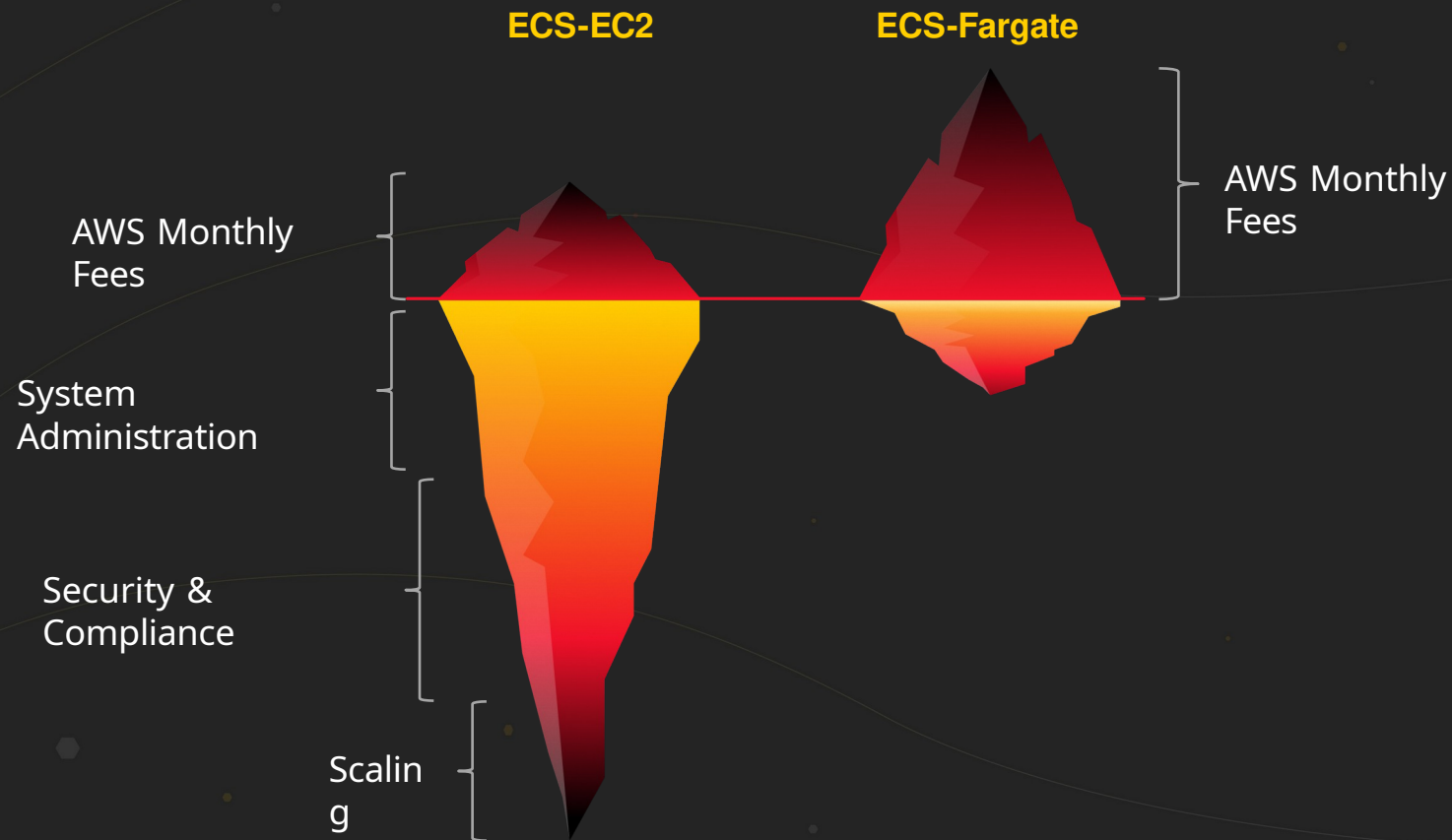


AWS Fargate! =



**Amazon
EC2**

Ops benefits



No more Amazon EC2 management



We no
longer
configure
AMIs



Monitoring
& logging is
built in

Operating system management



All of our code is packaged using Docker containers, so we are **ONLY** responsible for what runs within our containers

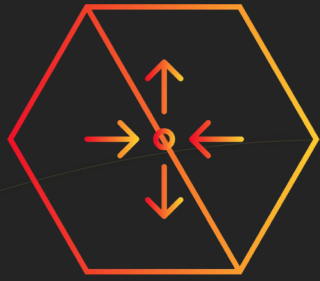
No more:

Linux
Patching

- Docker service updates

- ECS Agent updates

Scal g



We no longer deal
with scaling EC2
fleets



Taking care of bin
packing our
instance to run
cost-effectively

Compliance & security



Compliance and security are our top priorities

Out-of-the-box security as a
service. Fargate comes
certified with

Building an icecream service

- <http://icecream.datree.io>
- A simple Node.js app
- A web service using Koa.js
- Serving icecream!



Pipeline overview

- Source code in GitHub
- AWS Application Load Balancer
- AWS ECS Fargate cluster
- GitHub Actions workflow (CI/CD)

DEMO

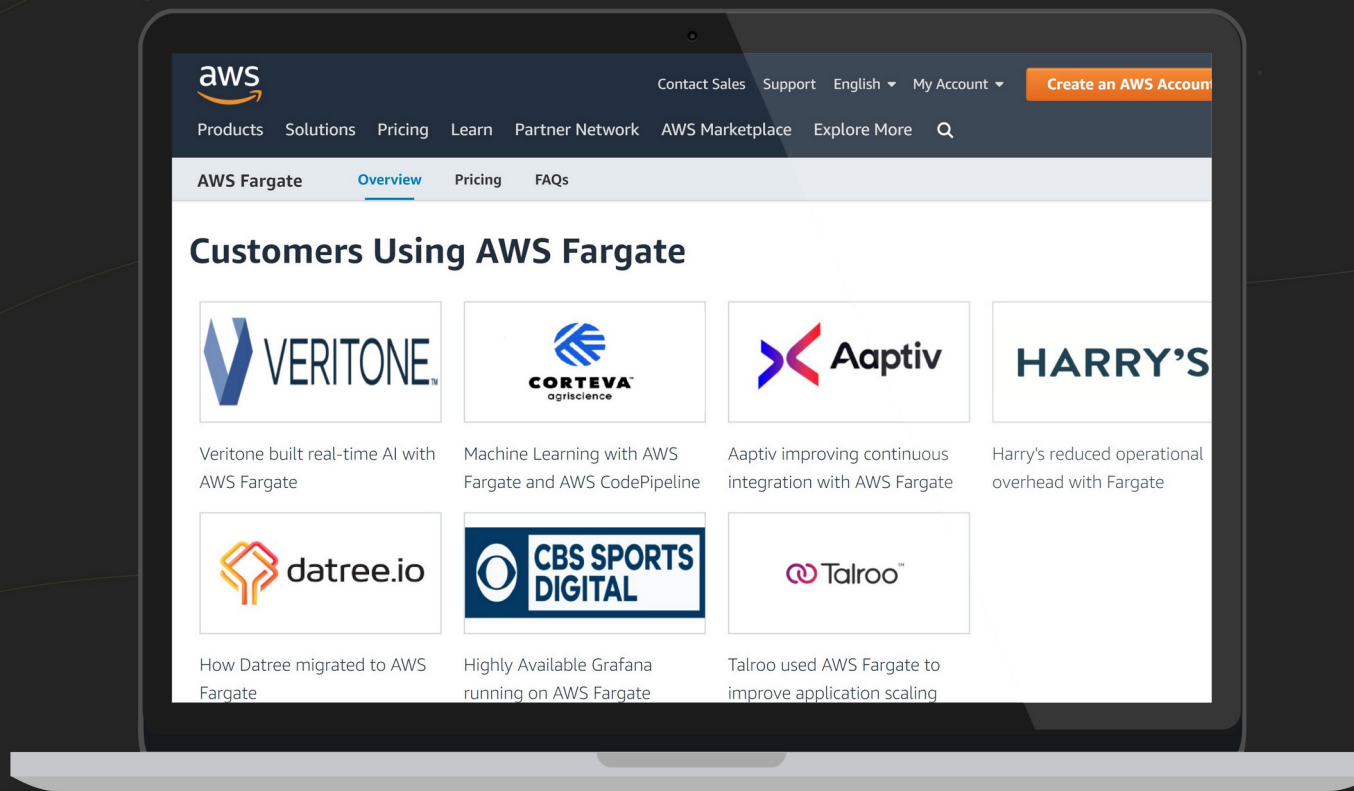
time

```
1  const Koa = require('koa')
2  const Router = require('koa-router')
3
4  const app = (module.exports = new Koa())
5
6  const router = new Router()
7
8  router.get('/health', ctx => {
9    ctx.body = { status: 'ice cream service is on!' }
10 })
11
12 router.get('/', ctx => {
13   ctx.body = { message: 'Welcome to the ice cream service!' }
14 })
15
16 router.get('/vanilla', ctx => {
17   ctx.body = { flavor: 'vanilla' }
18 })
19
20 router.get('/banana', ctx => {
21   ctx.body = { flavor: 'banana' }
22 })
23
24 app.use(router.routes()).use(router.allowedMethods())
25
26 console.log('Starting Koa server on port 8000')
27 if (!module.parent) app.listen(8000)
28
```


ECS on EC2 vs. Fargate

- 10 GB disk space limit
- No instance type selection (GPU/CPU/MEM optimized)
- Amazon EBS attaching is not available
- No Reserved Instance pricing
- No Spot Instances support

Open case study on AWS Fargate page



Resources

1. <https://datree.io/blog/migrating-to-aws-ecs-fargate-in-production/>
2. <https://www.youtube.com/watch?v=rtk3rRdAZ6s&feature=youtu.be&t=1239>
3. <https://github.com/silinternational/ecs-deploy>

Related breakouts

14:10 Deep Dive on Amazon Elastic Container Service (ECS) Brent Langston

15:00 Mastering Amazon Elastic Container Service for Kubernetes (Amazon EKS) Kobi Biton, Chen Fisher

15:50 From Code to a running container Alexei Ledenev, Gal Marder

Thank you!

Liron Dar

Dima Breydo

Shimon Tolts



<http://bit.ly/2SJ6Md2>